



Final Project Report:

FlavorSync – Advanced Digital Recipe Book.

Web Technologies – ICT 1209

Bachelor of Information and Communication Technology (Honors)

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Faculty of Technology

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Project Theme: Digital Recipe Book

Local Development URL: <http://localhost/ict-1209-miniproject-01/index.php>

GitHub Repository: <https://github.com/kavishka-disanayaka/ict-1209-miniproject-01.git>

Executive Summary

FlavorSync is an interactive, database-driven web application developed to help users find recipes, calculate portion sizes, and follow cooking instructions easily. Traditional web catalogs show static content that cannot adjust ingredient quantities or guide users during preparation. FlavorSync solves this by offering instant recipe filtering, client-side serving scaling, and a hands-free Cook Mode powered by the native browser Web Speech API.

The project is built entirely on the approved university stack: HTML5, CSS3, Bootstrap 5.3, Vanilla JavaScript, PHP 8, and MySQL via XAMPP. Server-side database operations use PHP Data Objects (PDO) with prepared statements to prevent SQL injection. User authentication is protected with salted Bcrypt password hashing.

- **Target Audience:** University students, home cooks, and busy individuals looking for quick meal instructions and adaptable serving sizes.
- **Core Value Proposition:** Intuitive recipe discovery, dynamic live portion scaling, and voice-assisted step-by-step guidance within a clean, responsive layout.

1. Introduction & Project Background

In modern culinary web applications, users are typically presented with fixed, non-responsive recipe instructions that assume static ingredient quantities and rely on manual keyboard/touch input during meal preparation.

FlavorSync addresses these bottlenecks through three technical innovations:

1. **Multi-Attribute Dynamic Filtering:** Instant client-side filtering by category, diet, cook time, and difficulty level without page reloads.
2. **Client-Side Mathematical Scaler:** Dynamically computing serving portions on-the-fly without incurring network latency or browser page refreshes.

3. **Hands-Free Auditory Interaction:** Utilizing the browser's native Web Speech API to narrate preparation steps so users with wet or messy hands do not touch their device screens.

2. Technology Stack & System Architecture

2.1 Technology Stack Details

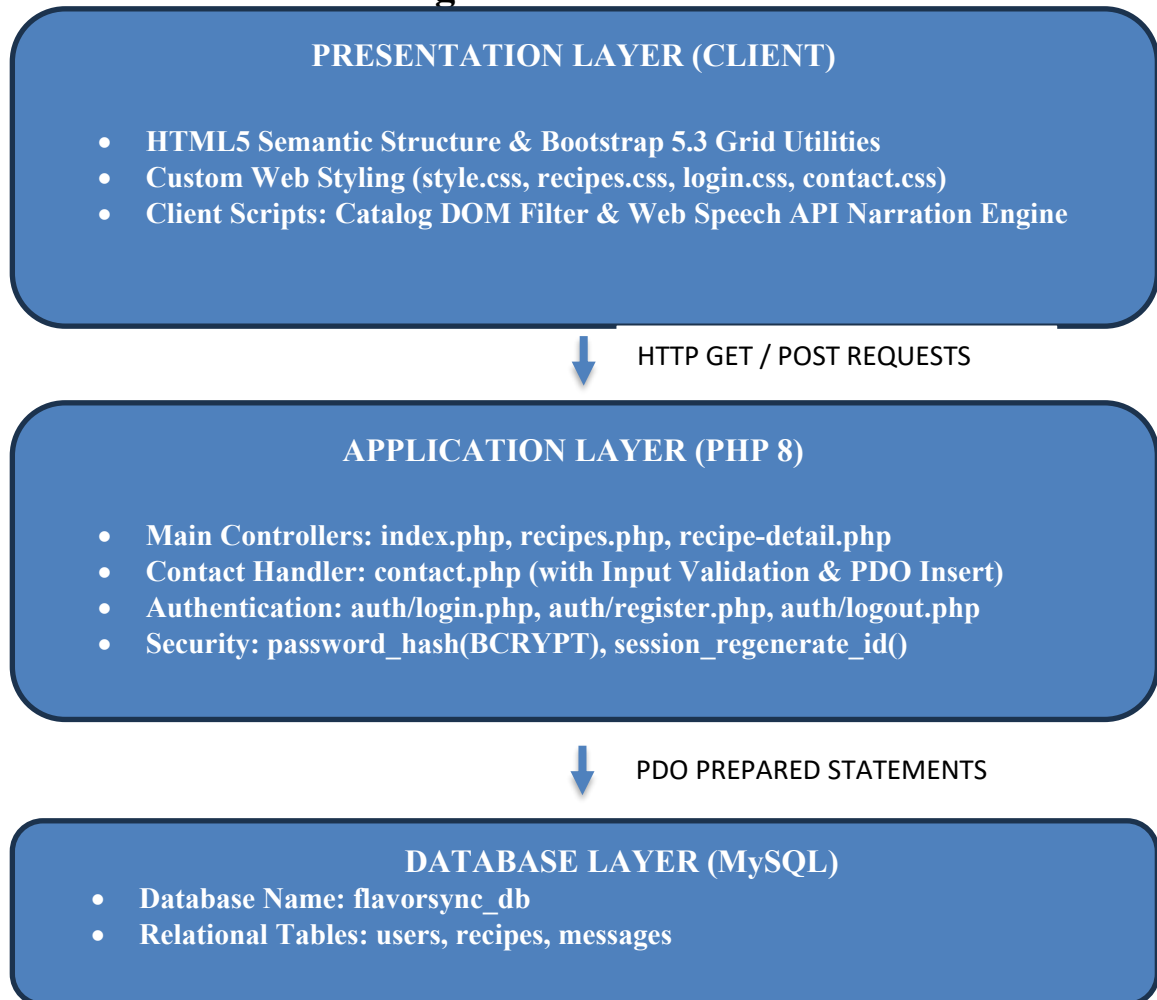
- **Frontend Structure:** Semantic HTML5 elements (<header>, <nav>, <main>, <section>, <article>, <footer>).
- **Styling & Layout:** CSS3 combined with Bootstrap 5.3.3 for a responsive grid layout, cards, and custom typography using Fraunces and Plus Jakarta Sans.
- **Client-Side Scripting:** Vanilla JavaScript (ES6+) for recipe card filtering, dynamic portion calculations, and browser text-to-speech audio narration.
- **Backend Engine:** PHP 8 procedural architecture handling sessions, form validations, and routing.
- **Database System:** MySQL (MariaDB 10.4.32 via XAMPP) running the InnoDB engine with UTF-8 character encoding
- **Version Control:** Git and GitHub for version tracking and team collaboration.

2.2 Technology Stack Matrix

Layer	Technology	Version/Tool	Purpose
Structure & Layout	HTML5	Living Standard	Semantic page layout, accessibility
Styling & UI	CSS3 & Bootstrap	Bootstrap v5.3	Responsive grid, utility classes, typography
Client Scripting	Vanilla JavaScript	ECMAScript 6+	Asynchronous event handling, DOM mutation, Web Speech API
Backend Logic	PHP	PHP 8.x	Server-side routing, database transactions, session control
Database	MySQL	InnoDB Engine	Relational persistence on XAMPP

Version Control	Git	GitHub	Version tracking, collaboration, release management
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2.3 Architectural Flow Diagram



2.4 Prescribed Folder Structure

```

ict-1209-miniproject-01/
├── asset/
├── auth/

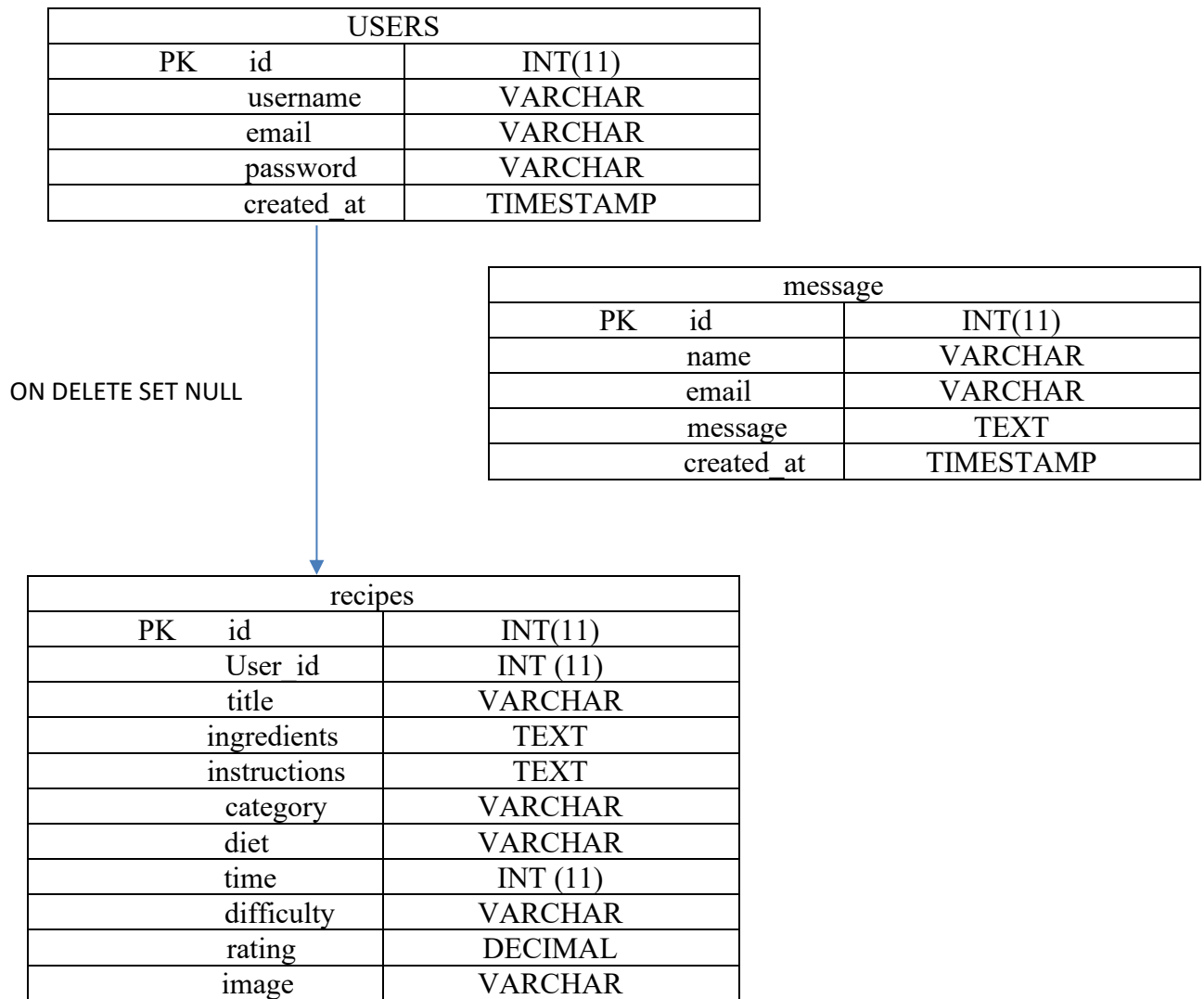
```

```
| |— login.php
| |— logout.php
| |— register.php
|— css/
| |— contact.css
| |— login.css
| |— recipes.css
| |— style.css
|— includes/
| |— db.php
|— js/
| |— contact.js
| |— recipes.js
| |— script.js
|— contact.php
|— flavorsync_db.sql
|— fridge-search.php
|— index.php
|— README.md
|— recipe-detail.php
|— recipes.php
```

3. Database Design & Implementation

The database connection is managed by includes/db.php using PDO with error reporting set to PDO::ERRMODE_EXCEPTION. The schema consists of three tables: users, recipes, and messages.

3.1 Entity Relationship Diagram



3.2 Data Dictionary

1.users Table

Stores registered user credentials.

- id (INT, Primary Key, Auto Increment): Unique identifier.
- username (VARCHAR(50)): User name.
- email (VARCHAR(100), Unique): User email address.
- password (VARCHAR(255)): Bcrypt-hashed password string.
- created_at (TIMESTAMP): Account registration timestamp.

2. recipes Table

Stores recipe content and categorization.

- id (INT, Primary Key, Auto Increment): Unique recipe identifier.
- title (VARCHAR(150)): Recipe name.
- ingredients (TEXT): Comma-separated ingredient list.
- instructions (TEXT): Step-by-step preparation method.
- user_id (INT, Foreign Key): References users.id.
- category (VARCHAR(50)): Meal type (breakfast, lunch, dinner, dessert).
- diet (VARCHAR(50)): Dietary category (veg, nonveg).
- time (INT): Cooking time in minutes.
- difficulty (VARCHAR(50)): Complexity level (easy, medium, hard).
- rating (DECIMAL(3,1)): Rating out of 5.
- image (VARCHAR(255)): Image path or URL.

3.messages Table

Stores messages sent through the contact form.

- id (INT, Primary Key, Auto Increment): Unique message ID.
- name (VARCHAR(100)): Sender name.
- email (VARCHAR(100)): Sender email.
- message (TEXT): Ingestion message body.
- created_at (TIMESTAMP): Submission timestamp.

4.System Implementation & Key Pages.

4.1Home Page(index.php)

- Features a Hero banner with direct search redirection to recipes.php.
- Includes dynamic Category filter chips (Breakfast, Lunch, Dinner, Dessert).
- Displays Popular Recipes cards and an interactive Trending Carousel built using Bootstrap 5.

4.2 Recipe Catalog & Filtering(recipes.php & recipes.js)

- Executes `SELECT * FROM recipes ORDER BY id DESC` via PDO to render all recipes dynamically.
- Implements client-side filtering using HTML5 data attributes (data-category, data-diet, data-time, data-difficulty).
- Allows instant filtering by meal type, dietary preference, prep time, and difficulty level without full-page refreshes.

4.3 Single Recipe & Cook Mode(recipe-detail.php)

- Reads the recipe dynamically using `recipe-detail.php?id=X` with parameterized PDO queries.
- **Live Serving Scaler:** Uses Vanilla JavaScript to increment and decrement serving counts cleanly:

```
let currentServings = 1;

function scaleIngredients(change) {

  if (currentServings + change >= 1) {

    currentServings += change;

    document.getElementById('servings-count').innerText = currentServings;

  }

}
```

- **Voice Cook Mode:** Uses browser Text-to-Speech via the Web Speech API to read instructions aloud:

```
function readAloud() {

  const text = document.getElementById('instruction-text').innerText;

  if ('speechSynthesis' in window) {

    window.speechSynthesis.cancel();

    const utterance = new SpeechSynthesisUtterance(text);

    utterance.rate = 0.9;

    window.speechSynthesis.speak(utterance);

  }

}
```

```
}  
  
}
```

4.4 User Authentication (auth/register.php & auth/login.php)

- **Registration:** Uses password_hash(\$password, PASSWORD_BCRYPT) to store secure hashed passwords in the database.
- **Login:** Uses password_verify(\$password, \$user['password']) to authenticate credentials.
- **Session Security:** Implements session_regenerate_id(true) upon successful authentication to safeguard against session fixation attacks.
- **Logout:** auth/logout.php clears and destroys sessions completely.

4.5 Contact Form & FAQ (contact.php)

- Displays studio contact metadata (*Mihintale, Anuradhapura*) and active contact channels.
- Inserts submitted feedback into the messages table using PDO prepared statements.
- Includes an accordion interface answering frequently asked user questions.

5. Security & Validation Practices

- (01) **SQL Injection Defense:** All SQL operations use PDO prepared statements with explicit parameter binding (:param), completely eliminating string concatenation.
- (02) **XSS Protection:** User-supplied output is escaped using htmlspecialchars() before rendering to HTML.
- (03) **Password Security:** Passwords are never stored in plain text; they are encrypted using one-way salted BCRYPT hashes.
- (04) **Input Validation:** Required input constraints are checked on both the client side (HTML5 required) and server side (trim(), !empty()).

6. Testing & Quality Verification

Test ID	Module	Test Scenario	Expected Outcome	Result
TC-01	Auth	Register user with valid details	User saved with \$2y\$10\$BCRYPT password hash	PASS

TC-02	Auth	Login with incorrect password	Login rejected; error alert shown	PASS
TC-03	Recipes	Click "Dessert" category filter	Only dessert recipe cards remain visible	PASS
TC-04	Detail	Click + on the serving scaler widget	Serving count increments without reloading	PASS
TC-05	Cook Mode	Click "Read Aloud" button	Browser reads cooking instructions aloud	PASS
TC-06	Contact	Submit valid contact inquiry	Message stored in messages table	PASS

7. Challenges and Solutions

1. Browser Speech Synthesis Overlap:

- *Problem:* Triggering the "Read Aloud" action multiple times in rapid succession caused speech streams to queue and overlap.
- *Solution:* Added `window.speechSynthesis.cancel()` right before initializing each new `SpeechSynthesisUtterance` instance to purge pending audio streams cleanly.

2. Preventing SQL Injection in Search and Dynamic Queries:

- *Problem:* Concatenating user inputs directly into SQL strings creates severe vulnerabilities.
- *Solution:* Replaced dynamic query string building with PDO prepared statements and explicit parameter binding (`bindParam`) throughout the codebase.

3. Session Hijacking and Fixation Defense:

- *Problem:* Session identifiers remaining static across authentication boundaries allow session fixation vulnerabilities.
- *Solution:* Implemented `session_regenerate_id(true)` immediately after successful authentication checks in `auth/login.php`.

8. Conclusion & Future Roadmap

FlavorSync fulfills all requirements for the ICT 1209 Web Technologies mini-project. The application integrates responsive frontend interfaces, clean client-side dynamic behaviors, and a secure PHP/MySQL database architecture.

8.1 Future Enhancements

- **User Recipe Creation:** Add frontend forms allowing registered users to insert their own recipes into the database.
- **Offline Recipe Storage:** Implement Progressive Web App (PWA) caching for reading recipes without an active internet connection.